AKSHAR VANDARA

Boston, MA | (857) 919-6234 | axr230102@gmail.com | linkedin.com/in/akshar-vandara | github.com/akshar-23 | portfolio

Software/Research Engineer • 3 peer-reviewed publications in PCG and constraint optimization • Shipped industry experience in Python, C++, C#, and full-stack systems

EDUCATION

Northeastern University — M.S., Game Science and Design *Boston, MA*
GPA: 3.9 / 4.0 | Coursework: Game Data Science, Game AI *May 2026*

Gujarat Technological University — B.E., Computer Engineering *Ahmedabad, India*
GPA: 3.6 / 4.0 | Coursework: Algorithms, AI, ML *May 2023*

PUBLICATIONS

- Vandara, A., Facey, K. and Cooper, S. 2025. Spacetime Level Generation and Editing with Constraints from Examples. *Proceedings of the AAAI Conference on Artificial Intelligence and Interactive Digital Entertainment*. 21, 1 (Nov. 2025), 142-152. DOI: https://doi.org/10.1609/aiide.v21i1.36818
- Vandara, A. and Cooper, S. 2026. Constraint-Based Four-Dimensional Spacetime Blending of Learned Game Mechanics Along Orthogonal Planes. *FDG Workshop on Procedural Content Generation, 2026* (accepted).
- Vandara, A. and Cooper, S. 2026. Spacetime Constraints for Continuous Gameplay. *FDG Workshop on Procedural Content Generation, 2026* (accepted).

EXPERIENCE

Research Assistant — Game Design Lab, Northeastern University (Boston, MA) *Feb 2025 - Present*
- Engineered constraint-solving pipeline in Python (Sturgeon system, Z3 SMT, PySAT) to generate 2D puzzle levels with provably valid solution paths, achieving **100–500× speedup** over prior academic methods (Space-Time WaveFunctionCollapse).
- First-authored peer-reviewed publication at **AAAI AIIDE 2025**; lead author on accepted **FDG 2026 PCG Workshop** paper extending the system to 4D spacetime blending of learned game mechanics across orthogonal planes (3D blended games from 2D inputs); lead author on accepted FDG 2026 PCG Workshop paper applying spacetime constraints to continuous, physics-based platformer gameplay using Z3 SMT.
- Built interactive editor for real-time spacetime editing of generated levels and solution paths; integrated VTK/ParaView visualization for 3D voxel output.

Gameplay Systems & UI Programmer — Otisco Studios (Orlando, FL) *May 2025 - Aug 2025*
- Built **8+ core gameplay systems** across two titles, including save/load, environment fracturing, health/score management, UI menus, and dynamic spawning systems.
- Implemented **5+ enemy and NPC AI archetypes** using Unreal Behavior Trees, Blackboards, and Environment Query System (EQS) for roguelike and open-world combat.
- Resolved 30+ critical pre-launch bugs and refactored codebases across multiple projects, doubling average gameplay framerate.

Unity Developer — Arcadon Gamelab (Bengaluru, India) *Mar 2023 - Jul 2024*
- Architected sports management backend handling **1,000+ concurrent users** via Firebase Realtime DB and Cloud Functions with sub-100ms response times for pseudo-multiplayer features.
- Built simulation engine processing **30 matches/min**; reduced load times 40% and improved gameplay responsiveness 30% via state-machine refactor. Led alpha testing with 100+ users across 3 release iterations, reaching 85% satisfaction.

PROJECTS

Spacetime PCG Research Suite (GitHub / AIIDE Publication) *Python, Z3 SMT, PySAT, Pandas, ParaView*
- Open-source extension of Sturgeon for spacetime-constrained level generation; supports 2D, 3D blended, and continuous physics-based platformer domains.

Red Circuit Rebellion (Steam release) *Unreal Engine 5, C++, Blueprints, Behavior Trees*
- Shipped mech-destruction action game with 5+ core systems (save/load, environment fracturing, health, objectives), 3 EQS-driven enemy archetypes, and 2 game modes (Story, Arcade). Doubled average FPS via blueprint and asset optimization.

Cricket Manager: League (Google Play release) *Unity, C#, Firebase, Cloud Functions*
- Sports management title with cloud-synced inventory, AI opponent simulation, and live ops backend supporting 1,000+ users; published to Play Store with 85% alpha satisfaction.

SKILLS

- **Languages:** Python, C++, C#, JavaScript/TypeScript, Java, SQL, R, C
- **ML / AI:** PyTorch, TensorFlow, NumPy, Pandas, CUDA, Z3 SMT, PySAT, constraint optimization, SAT/CSP modeling, OpenCV
- **Backend / Web:** Flask, Django, SQLAlchemy, Node.js, React, Next.js, REST APIs, Firebase Realtime DB, Firebase Cloud Functions
- **Game Dev & Tools:** Unity (2D/3D), Unreal Engine 5, Blueprints, Behavior Trees, EQS, Git, Perforce, Visual Studio, ParaView/VTK